Implementing SSH Brute-Force Simulation, Log Analysis, and System Hardening on Windows

Capstone Security Lab Report

Submitted By

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Abstract

This project focuses on identifying and defending against SSH brute-force attacks on a Windows Server running OpenSSH. The lab simulates a real-world cyberattack using a Kali Linux machine and demonstrates how security analysts can investigate authentication logs, determine whether unauthorized access occurred, and apply defensive security controls.

The project covers network verification, SSH configuration, attack simulation with Hydra, Windows Security log analysis, incident assessment, and system hardening. Security improvements such as limiting login attempts, disabling password-based authentication, restricting SSH users, and configuring firewall rules were implemented to reduce the likelihood of future attacks.

This lab demonstrates practical incident response and defensive security skills commonly used in modern enterprise environments.

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1. Introduction

Remote administration services such as SSH are frequently targeted by attackers attempting to gain unauthorized access to systems. One common attack method is brute-force authentication, where an attacker repeatedly tests username and password combinations until valid credentials are discovered.

In this project, a Windows Server configured with OpenSSH was intentionally exposed to a simulated brute-force attack launched from a Kali Linux machine. The objective was not only to observe how the attack behaves but also to investigate how Windows records authentication events and how defensive measures can be implemented to secure the system.

The project replicates tasks performed by cybersecurity professionals and SOC analysts, including attack detection, log investigation, incident analysis, and defensive configuration.

2. Project Objectives

The goals of this project were:

- Configure and verify SSH functionality on Windows Server.

- Simulate an SSH brute-force attack using Hydra.
- Examine Windows Security logs for authentication activity.
- Identify failed and successful login attempts.
- Determine whether the attack resulted in system compromise.
- Apply security hardening measures to protect the SSH service.
- Restrict unauthorized remote access using firewall rules.
- Improve monitoring and incident response readiness.

3. Lab Environment

The project was conducted in a controlled virtual lab environment.

System	Role	Operating System
Windows Machine	Target Server	Windows Server 2019/2022
Kali Linux	Attacker Machine	Kali Linux

Network Setup

Device	IP Address
Windows Server	10.0.0.5
Kali Linux	10.0.0.6

Both systems were connected to the same internal network to allow communication between the attacker and target machines.

4. Software and Tools Used

Tool	Purpose
OpenSSH Server	Remote SSH access
PowerShell	Windows administration
Hydra	Brute-force attack simulation
Event Viewer	Authentication log analysis

Windows Defender Firewall	Network access restriction
Kali Linux	Penetration testing environment

5. Methodology

The project was divided into six phases:

1. Verify network and SSH functionality
2. Create user accounts for testing
3. Simulate a brute-force attack
4. Analyze security logs
5. Evaluate compromise status
6. Harden the system against attacks

Each phase was completed sequentially to ensure accurate testing and analysis.

6. Phase 1 – Initial System Verification

6.1 Verifying Network Connectivity

The first task involved testing communication between the Kali Linux system and the Windows Server using the ping utility.

Command

```
ping 10.0.0.5
```

Purpose

This test confirmed that both systems could communicate across the network before SSH testing began.

Result

The Windows machine responded successfully, confirming that network connectivity was functioning correctly.

6.2 Verifying OpenSSH Installation

PowerShell was opened with administrator privileges to verify whether the OpenSSH server feature was installed.

Command

```
Get-WindowsCapability -Online | Where-Object Name -like 'OpenSSH*'
```

Result

The output confirmed that the OpenSSH Server component was not installed on the Windows machine.

Hence, I run the below command to get the service installation.

```
Add-WindowsCapability -Online -Name OpenSSH.Server~~~~0.0.1.0
```

6.3 Starting the SSH Service

The SSH daemon was started and configured to run automatically at startup.

Commands

```
Start-Service sshd
Set-Service -Name sshd -StartupType Automatic
```

Result

The SSH service started successfully without errors.

6.4 Configuring Firewall Access

A firewall rule was created to allow inbound SSH traffic through TCP port 22.

Command

```
New-NetFirewallRule -Name sshd -DisplayName "OpenSSH Server" -
Enabled True -Direction Inbound -Protocol TCP -Action Allow -
LocalPort 22
```

Result

The firewall rule allowed remote SSH connections to the Windows Server.

7. Phase 2 – User Account Creation and SSH Testing

7.1 Creating Test Accounts

Two local user accounts were created for authentication testing.

Commands

```
net user testuser1 Password123! /add
net user adminuser SecurePass! /add
```

Verification

The accounts were confirmed using:

```
net user
```

7.2 Testing SSH Authentication

SSH login testing was performed from the Kali Linux machine.

Command

```
ssh testuser1@<Windows_IP>
```

Result

After entering the correct password, the connection successfully opened a remote Windows command-line session.

This confirmed that the SSH server was functioning properly before the attack simulation.

8. Phase 3 – SSH Brute-Force Attack Simulation

8.1 Preparing the Password List

The available wordlists on Kali Linux were checked.

Command

```
ls /usr/share/wordlists/
```

The commonly used `rockyou.txt.gz` password dictionary was selected for testing.

As it is compressed, it was extracted using:

Command

```
Sudo gunzip /usr/share/wordlists/rockyou.txt.gz
```

8.2 Launching the Brute-Force Attack

Hydra was used to automate repeated SSH login attempts.

Firstly, checked whether the Hydra is installed on the Linux using the below command

Command

hydra -h

Command

```
hydra -l testuser1 -P /usr/share/wordlists/rockyou.txt
ssh://10.0.0.5
```

Attack Activity Observed

During the attack:

- Hundreds of password attempts were generated.
- Multiple authentication failures occurred.
- Windows Security logs recorded each failed login attempt.
- Hydra continued cycling through passwords until manually stopped.

The attack was terminated after several minutes using:

```
Ctrl + C
```

9. Phase 4 – Security Log Investigation

9.1 Opening Event Viewer

Windows Event Viewer was used to inspect authentication activity.

Navigation Path

```
Event Viewer → Windows Logs → Security
```

9.2 Investigating Failed Login Events

The Security log was filtered using Event ID 4625.

Meaning of Event ID 4625

This event identifies failed login attempts.

Information Collected

Each event contained:

- Username used
- Source IP address
- Authentication type
- Timestamp
- Failure details

Observation

A large number of failed authentication attempts originated from the Kali Linux IP address, confirming the brute-force activity.

9.3 Investigating Successful Login Events

The logs were filtered again using Event ID 4624.

Meaning of Event ID 4624

This event indicates a successful login.

Observation

Only legitimate login activity performed during SSH testing was observed. No unauthorized successful login events were identified during the attack simulation.

10. Phase 5 – Compromise Evaluation

The next step involved determining whether the attacker gained access to the system.

The following questions were investigated:

- Did any successful logins occur during the attack period?
- Were any valid accounts compromised?
- Did successful login timestamps match the Hydra attack?

Analysis

Although many failed authentication attempts were recorded, no evidence suggested that the brute-force attack successfully compromised the system.

Conclusion

The attack generated suspicious activity but did not result in unauthorized access.

11. Phase 6 – SSH Hardening and System Defense

After analyzing the attack, several defensive security measures were implemented.

11.1 Disabling Password Authentication

The SSH configuration file was opened:

```
notepad C:\ProgramData\ssh\sshd_config
```

The following setting was modified:

Before

```
#PasswordAuthentication yes
```

After

```
PasswordAuthentication no
```

Purpose

This change forced users to authenticate using SSH keys instead of passwords, reducing exposure to brute-force attacks.

11.2 Limiting Login Attempts

The following setting was added:

```
MaxAuthTries 3
```

Purpose

Restricting login attempts helps reduce automated password-guessing attacks.

11.3 Restricting SSH Access to Authorized Users

Only approved accounts were allowed to connect using SSH.

Configuration

```
AllowUsers adminuser
```

Purpose

This prevented unauthorized or unnecessary accounts from using SSH.

11.4 Restarting the SSH Service

The SSH service was restarted to apply configuration changes.

Command

```
Restart-Service sshd
```

11.5 Restricting SSH Access by IP Address

Firewall restrictions were configured to allow SSH access only from trusted systems.

Command

```
Set-NetFirewallRule -Name sshd -RemoteAddress 10.0.0.6
```

Purpose

This reduced the attack surface by limiting remote access to specific IP addresses.

12. Final Security Validation

After implementing security controls, the Hydra brute-force attack was repeated.

Result

The hardened configuration significantly reduced attack effectiveness:

- Unauthorized access attempts failed faster.
- Password-based attacks became ineffective.
- SSH access was limited to authorized users and trusted IPs.

The defensive measures successfully improved the security posture of the system.

13. Results and Observations

Security Task	Result
SSH service configured successfully	Yes
Brute-force attack simulated	Yes
Failed login events detected	Yes
Successful unauthorized login detected	No
Security hardening completed	Yes
Firewall restrictions applied	Yes

14. Challenges Encountered

Several technical challenges occurred during the lab:

- Initial SSH connection troubleshooting
- Verifying correct firewall configuration
- Handling large volumes of security log entries
- Ensuring SSH service changes applied correctly
- Differentiating legitimate logins from attack attempts

These issues were resolved through configuration review and repeated testing.

15. Conclusion

This project demonstrated how SSH brute-force attacks can target Windows systems running OpenSSH and how security analysts can respond effectively using log analysis and defensive hardening techniques.

By simulating an attack and reviewing Windows Security logs, it was possible to identify authentication attempts, analyze suspicious activity, and determine whether unauthorized access occurred. Additional hardening measures such as disabling password authentication, restricting users, limiting login attempts, and applying firewall rules significantly improved overall security.

The lab provided valuable practical experience in attack detection, incident investigation, and system defense.

16. Recommendations

The following improvements are recommended for future security enhancement:

- Implement multi-factor authentication (MFA)
- Deploy centralized log monitoring solutions
- Configure automated alerting systems

- Apply account lockout policies
- Regularly review authentication logs
- Use stronger password policies
- Conduct periodic penetration testing

17. References

1. Microsoft OpenSSH Documentation
2. Microsoft Event Viewer Documentation
3. Hydra Penetration Testing Tool Documentation
4. Windows Defender Firewall Documentation
5. NIST Cybersecurity Framework
6. CIS Security Benchmarks for Windows Server